Captain’s Summer Exchange - Montgomery High School - Repaired practice edition - Answer key

Division C | Codebusters | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

repair-case. (6 points) Authored written substitute for unavailable source material: Solve this case and explain your reasoning: Puzzle A offers 100 points with a 0.9 estimated solve probability in 3 minutes; B offers 200 with 0.3 probability in 6 minutes. Compare expected points per minute as a planning aid.

reasoning: A gives 100 x 0.9 / 3 = 30 expected points per minute.
B gives 200 x 0.3 / 6 = 10 expected points per minute.
Estimates are uncertain; revise them if new constraints make B easier or actual scoring differs. (3 points).

conclusion: A is a reasonable initial priority under these assumptions; the ratio is a planning heuristic, not a score guarantee. (3 points).